Pre Lab: Calibrating a Function Generator

The Fourier Transform

For a signal $x(t)$, the continuous time Fourier Transform $X(\omega)$ can be found as

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

The Fourier Transform is a mathematical tool used to decompose a time-based signal into its individual frequency components.

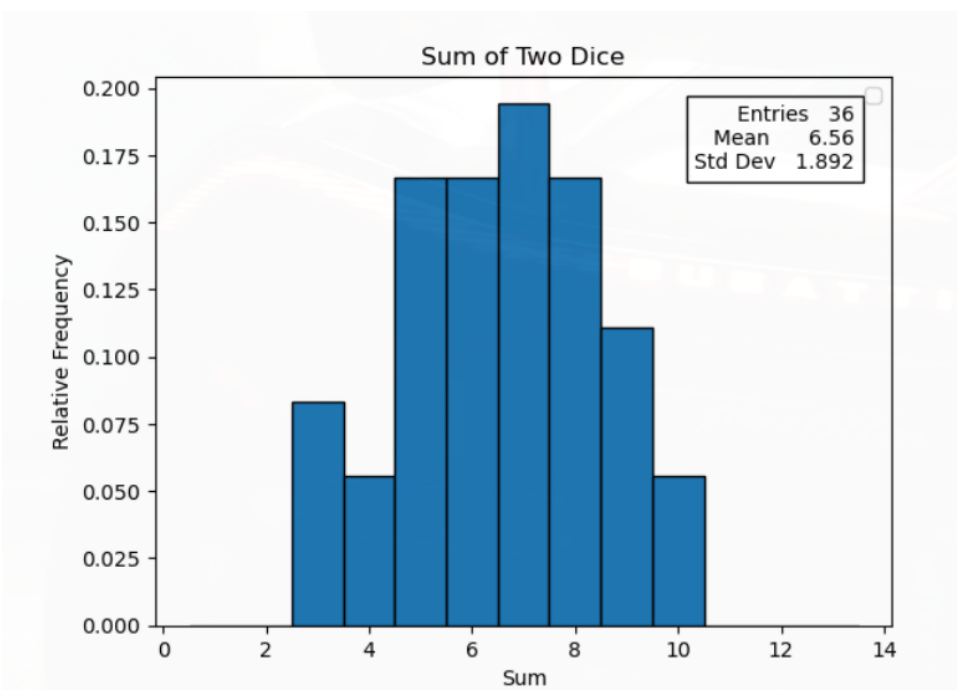
Comparing Measurements

Measurement and Uncertainty: Problem 14

Part A: Problem 8 – Dice Simulation vs Perfect Distribution

For the theoretical values for a two-dice roll, the exact mean is $\mu = 7$ and the standard deviation is $\sigma = \sqrt{6} \approx 2.45$. Because these are theoretical, their uncertainty is 0.

In my simulation data:



I get a mean of $\bar{x} = 6.56$ and a standard deviation of $s_x = 1.892$.

1. *Comparing the means*

The uncertainty of the simulated means is $s_{\bar{x}} = \frac{s_x}{\sqrt{N}} = \frac{1.892}{\sqrt{36}} = 0.315$

Next, calculating the t-value for the difference between the simulated mean and the theoretical mean:

$$t_{\text{mean}} = \frac{|6.56 - 7|}{\sqrt{0.315^2 + 0^2}} = \frac{0.44}{0.315} = 1.40$$

According to the Normal (Gaussian) probability table, a t-value of 1.40 falls between 1.0 (68% probability of difference) and 1.5 (87% probability of difference). Because the t-value is relatively low, we cannot definitively say the simulated mean is significantly different from the theoretical mean.

2. *Comparing the Standard Deviations*

Calculating the fractional uncertainty of the simulated standard deviation using $N = 36$

$$s_{s_x} = \frac{s_x}{\sqrt{2(N-2)}} = \frac{1.892}{\sqrt{2(34)}} \approx 0.229$$

Calculating the t-value for the standard deviations:

$$t_{std} = \frac{|1.892 - 2.45|}{\sqrt{0.229^2 + 0^2}} = \frac{0.558}{0.229} \approx 2.44$$

Referring again to the Normal probability table, a t-value of 2.44 inficates a roughly 98% to 99% probability that the difference is significant.

Conclusion: While the simulated mean matches the expectation reasonably well, the standard deviation of the simulated data is significantly tighter than the theoretical prediction.

Part B: Problem 10b – Reaction Time vs. Paper Fall Spread

Finding the differences between the standard deviation between the reaction time and paper fall experiments.

- Reaction time: $s_1 = 0.120s$ for $N_1 = 20$ trials
  - The uncertainty of s_1 is found by $u(s_1) = \frac{0.120}{\sqrt{2(18)}} = 0.020s$
- Paper fall: $s_2 = 0.241s$ for $N_2 = 20$ trials
  - Uncertainty of s_2 is found by $u(s_2) = \frac{0.241}{\sqrt{2(18)}} \approx 0.040s$

Calculate the combined uncertainty of the difference and the resulting t-value:

$$t = \frac{|s_2 - s_1|}{\sqrt{u(s_1)^2 + u(s_2)\text{^}2}} = \frac{|0.241 - 0.120|}{\sqrt{0.020^2 + 0.040^2}} = \frac{0.121}{0.0447} \approx 2.71$$

Conclusion: A t-value of 2.71 corresponds to a >99% probability of a difference. This proves quantitively that the physical mechanics of dropping a piece of paper introduces a significantly larger statistical spread than human reaction time alone.

Problem 15: Ring Measurement and Uncertainty Propagation

- Rind Diameter (D): 20.0 mm
- Circumference (C): 62.5 mm (measured by wrapping a string, then measuring the string)

1. Defining the Uncertainties:

I am using an analog metric ruler with 1 mm marks. The limit of precision (half-width a) is 0.5 mm. Assuming a triangular PDF for reading an analog scale, the Type B uncertainty for both measurements is:

$$s_C = s_D = \frac{a}{\sqrt{6}} = \frac{0.5}{\sqrt{6}} \approx 0.204 \text{ mm}$$

2. Propagating the Uncertainty of the Ratio

Finding the ratio $z = \frac{C}{D}$ evaluates to 3.125

Because the same ruler was used for both measurements, the Type B scale uncertaintis are perfectly correlated, meaning the correlation coefficient $r_{CD} = +1$.

Applying the general uncertainty propagation formula (equation 12):

$$s_z^2 = \left(\frac{\partial z}{\partial C}\right)^2 s_C^2 + \left(\frac{\partial z}{\partial D}\right)^2 s_D^2 + 2\left(\frac{\partial z}{\partial C}\right)\left(\frac{\partial z}{\partial D}\right) r_{CD} s_C s_D$$

Calculating the sensitivity coefficients:

- $\frac{\partial z}{\partial C} = \frac{1}{D} = \frac{1}{20.0} = 0.05 \text{ mm}^{-1}$
- $\frac{\partial z}{\partial D} = -\frac{C}{D^2} = -\frac{62.5}{400.0} = -0.156 \text{ mm}^{-1}$

Plug into the variance equation:

$$s_z^2 = (0.05)^2(0.204)^2 + (-0.156)^2(0.204)^2 + 2(0.05)(-0.156)(1)(0.204)^2$$

$$s_z^2 = (0.204)^2\,[0.0025 + 0.0243 - 0.0156] = (0.0416) \times (0.0112) \approx 0.000466$$

$$s_z = \sqrt{0.000466} \approx 0.022$$

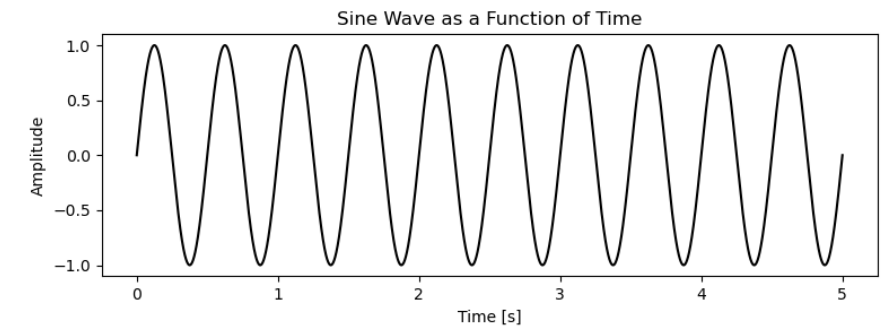
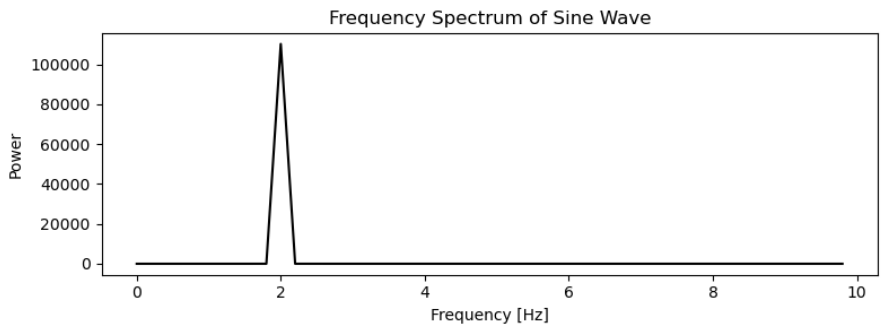
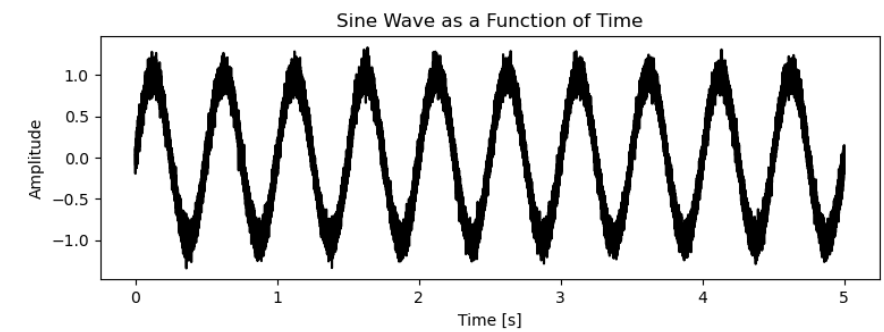
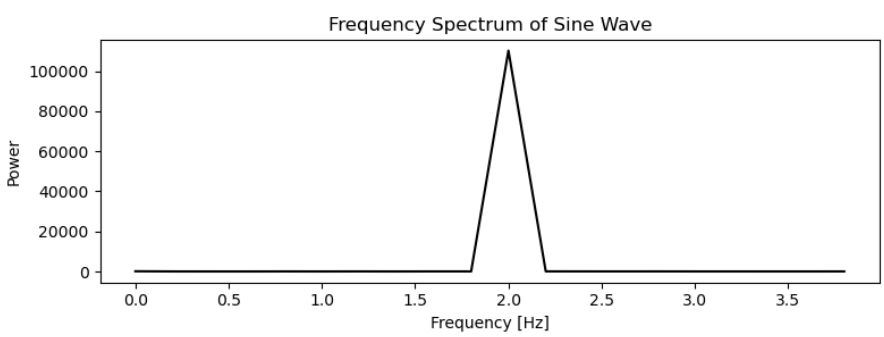
The final Measurement for the ratio is 3.125 ± 0.022.

3. Comparison to Expectation:

The expected ratio of circumference to diameter is π. The significance of the difference can be checked using a t-test:

$$t = \frac{|\pi - z|}{s_z} = \frac{|3.142 - 3.125|}{0.022} = \frac{0.017}{0.022} \approx 0.77$$

Conclusion: A t-value of 0.77 indicates a roughly 50% probability of difference, which is well below the threshold for significance. Therefore, the simulated measurements agree with the geometric expectation of π.

Graphing Sine Waves with Python

With inputs of $f = 2\ \mathrm{Hz}$ and $t = 5\ \mathrm{s}$ for both